Lab 7: Fuzzing for Input Validation Bugs

Fuzz testing with Hypothesis

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SRN	PES1UG24CS251 (5th Sem, Section E)
Repository	L6_L7_06_event_ticket_tier (Event Ticket Tier), folder: Lab 7
Module	processor.py: decode_promo_voucher, calculate_seat_tier_price, parse_attendee_badge
Tools	Python 3.14, pytest 9.2.0, Hypothesis 6.170.15

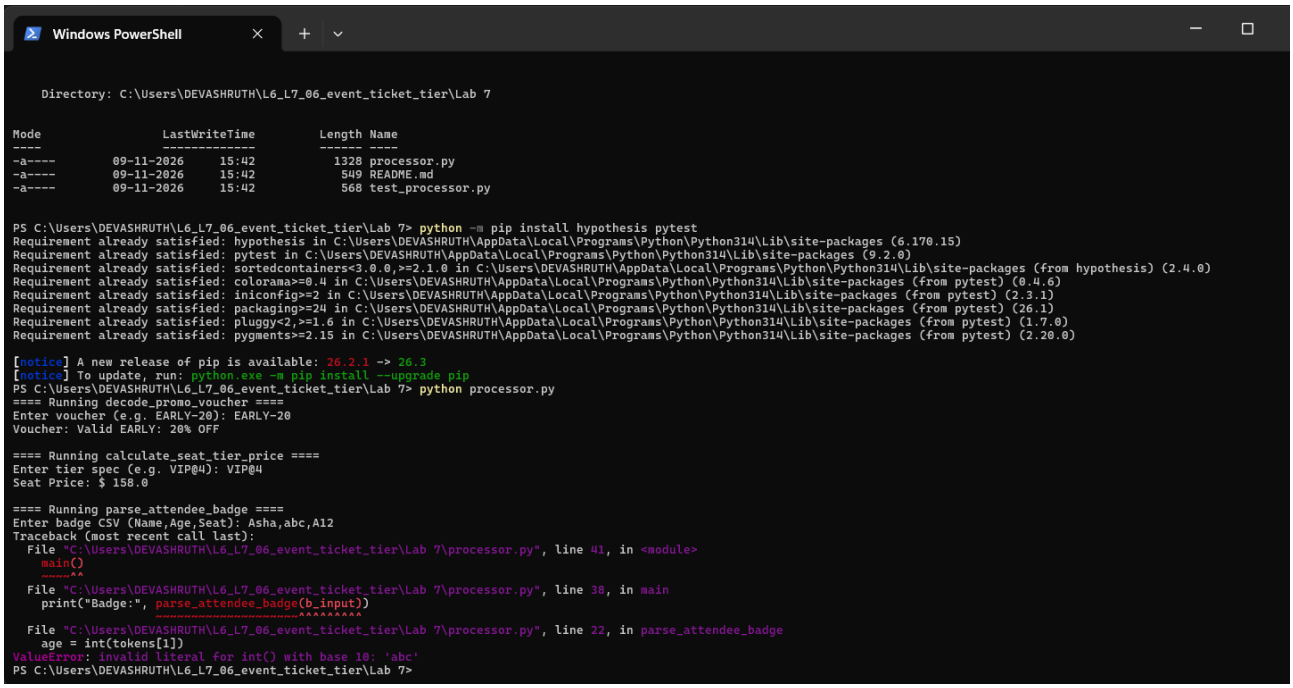
Bugs found by fuzzing and how they were fixed

Function	Bugs found	Fix
decode_promo_voucher	Crash on None; crash when the code has no dash or more than one dash; crash on a non-numeric percentage; accepted impossible discounts such as 101%.	Type check, exactly one '-' separator, ASCII-letter/digit prefix, whole-number percentage limited to 0-100; otherwise returns 'Invalid voucher: ...'.
calculate_seat_tier_price	Crash on None; crash when '@' is missing; crash on a non-integer row; KeyError for an unknown tier; negative rows gave a price below the cheapest tier; huge rows gave absurd prices.	Type check, exactly one '@', tier must be VIP/PREMIUM/GENERAL, row a plain whole number from 0 to 1000; otherwise returns None (main() prints a friendly message).
parse_attendee_badge	Crash on None; IndexError when fewer than 3 fields; crash on a non-numeric age; accepted negative or impossible ages.	Type check, exactly 3 fields, non-empty name and seat, age a whole number from 0 to 120; otherwise returns 'Invalid badge: ...'.

Tests: three Hypothesis 'must never crash' tests (arbitrary text or None) plus three structured tests that check the results make sense (discount 0-100, price at least the cheapest tier, age 0-120).

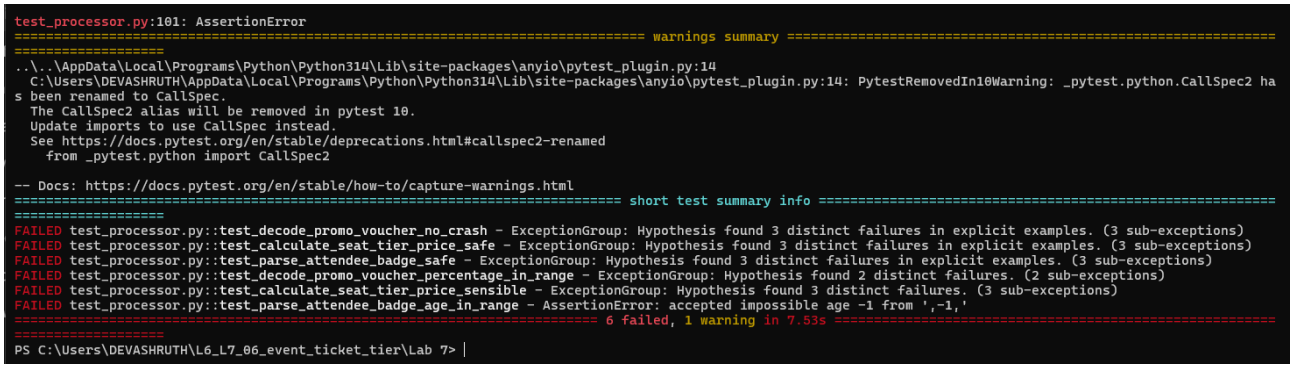
Screenshot 1: Step 2 - Run the buggy processor.py

python processor.py with the original code. The first two functions work on valid input, but parse_attendee_badge crashes on the badge Asha,abc,A12 with ValueError: invalid literal for int() with base 10: 'abc'.



Screenshot 2: Step 4 - Hypothesis tests failing on the buggy code

python -m pytest test_processor.py against the original processor.py: all 6 tests fail. The fuzzing found crashes on None, missing separators and non-numeric values, and logic bugs that do not crash (a 101% discount, a seat price below the cheapest tier, an age of -1).



Screenshot 3: Step 5 - Run the fixed processor.py

python processor.py with the hardened code and the same inputs as Step 2. The invalid badge now returns 'Invalid badge: age must be a whole number' instead of crashing.



Screenshot 4: Step 6 - Hypothesis tests passing on the fixed code

python -m pytest test_processor.py -v against the fixed processor.py: all 6 tests pass (6 passed). The top of the screen also shows the fixed script running again.

